

DATA ANALYSIS & STRATEGIC INSIGHTS REPORT

Project Theme: Retail Customer Loyalty, Behavioral Segmentation & Revenue Optimization

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Tools Used: SQL Server / PostgreSQL, Microsoft Power BI Desktop

1. Executive Summary

This report provides a data-driven breakdown of purchase patterns, promotional habits, and behavioral segmentations for a retail database consisting of **4,000 unique customers**. The main objective of this analysis is to evaluate whether business revenue is driven by genuine, organic brand pull or a high reliance on marketing discounts. By engineering custom segments in SQL and translating the metrics into dynamic Power BI visualizations, this report identifies high-value regional hubs, seasonal trends, and product-specific cross-selling funnels to maximize profit margins.

2. Key Performance Indicators (KPIs)

The top-tier card visuals in the dashboard provide a clear health check of overall retail performance:

- **Total Customer Base: 4,000** Unique Customers active within the transactional database.
- **Core Retention Rate: 39.21%** — Indicating that nearly 40% of the customer base returns for repeat purchases, showing a healthy foundation of baseline loyalty.
- **Average Order Value (AOV): 60** — The average gross spending power per individual transaction across all product classes.
- **Loyal Customer Rating: 3.77 out of 5.0** — The baseline satisfaction score calculated across retained customer brackets.

3. SQL Behavioral Segmentation Logic

To create actionable segments, custom data-engineering scripts were implemented at the backend database layer. Customers were categorized into five highly specific tiers based on their promotion application history and purchase frequencies:

SQL

CASE

WHEN discount_applied = 'Yes' AND previous_purchases <= 15 THEN 'Bargain Hunter'

WHEN discount_applied = 'Yes' AND previous_purchases BETWEEN 16 AND 30 THEN 'Opportunistic Buyer'

WHEN discount_applied = 'Yes' AND previous_purchases > 30 THEN 'Promo-Loving Loyal'

WHEN discount_applied = 'No' AND previous_purchases <= 15 THEN 'New Organic Customer'

```
WHEN discount_applied = 'No' AND previous_purchases > 15 THEN 'Pure Genuinely Loyal'  
ELSE 'Other'  
END AS Customer_Segment
```

Transactional Value Brackets

Gross spending amounts for individual item purchases were partitioned into simple tiers to quickly isolate consumer purchasing power:

- **Low Spend:** le 40 per purchase.
- **Medium Spend:** Between 41 and 70 per purchase.
- **High Spend:** >70 per purchase.

4. Deep-Dive Data Model Insights

A. Customer Value Pyramid & Segments

According to backend execution matrices, **Pure Genuinely Loyal** and **New Organic Customers** generate the healthiest, highest-margin revenue because they pull value without relying on price drops. Conversely, **Bargain Hunters** exhibit high transaction frequencies exclusively when promotions are active, yielding tight unit margins.

B. Category Demand Funnel

- **Volume Distribution: Clothing (1.74K Orders) and Accessories (1.24K Orders)** vastly outpace *Footwear (0.60K)* and *Outerwear (0.32K)* in overall transactional volume.
- **Funnel Logic Filter:** When analyzing purchase history cohorts, a clear lifecycle pattern emerges:
 - *Low Purchase History Customers* use **Accessories** and **Footwear** as entry-level gateway products.
 - *High Purchase History Customers* scale their lifetime value into higher-end core categories such as **Clothing** and specialized seasonal **Outerwear**.

C. Promotional Engagement & Demographics

- **Gender-Wise Split:** Female consumer segments utilize and react to promotional codes at a significantly higher rate (~68%) compared to male segments (~32%), making targeted marketing highly distinct between groups.
- **VIP Customer Profile Extraction:** Filtering for highly loyal customers who buy completely without discounts (`discount_applied = 'No'` and `previous_purchases >= 40`) isolates a premium subset displaying specific operational traits:
 - **Average Age:** Consistently trends around mature spending brackets.
 - **Satisfaction:** Generates high review scores (≥ 4.0).

- **Logistics Preferences:** Heavily favors **Credit Cards** or secure digital wallets for transactions, combined with **Express/Premium Shipping** fulfillment.

5. Dashboard Architecture & Visual Analysis

Visual 1: Revenue Contribution by Customer Segment (Top-Left Funnel)

- **Description:** An inverted funnel structure mapping customer tier counts.
- **Key Finding: Pure Genuinely Loyal** buyers form the wide, stable top tier of the brand value pyramid. They act as the primary engine for organic growth and must be insulated from competitive churn.

Visual 2: Retention Rate vs. Promotional Dependency (Central Scatter Plot)

- **Description:** A behavioral quadrant matrix plotting promotion reliance (X-axis) against retention probability (Y-axis).
- **Key Finding:** It clearly visualizes that high discount reliance does not automatically create long-term loyalty. The cohorts with low promo dependency maintain standard, resilient retention curves, proving high inherent brand equity.

Visual 3: Geographic Opportunity Mapping (Bubble Map Visual)

- **Description:** A geographical coordinate map where bubble diameters represent Total_Spend volumes and color legends categorize customer types.
- **Key Finding:** This map allows marketing leads to easily pinpoint regions displaying a premium combination: dense clusters of high spending power coupled with low promo dependency. These locations represent immediate expansion targets for high-margin retail pushes.

6. Strategic Business Recommendations

1. **Protect Core Organic Margins:** Avoid offering flat product discounts to the **Pure Genuinely Loyal** segment. They have strong brand affinity and respond better to value-added experiences like early access tokens, premium loyalty points, community rewards, or dedicated client care lines.
2. **Optimize the Cross-Selling Funnel:** Build an automated marketing workflow that targets new organic buyers right after they purchase an entry product like an *Accessory*. Use targeted recommendations to guide them toward higher-value categories like *Clothing* and *Outerwear*, accelerating their journey up the loyalty ladder.
3. **Deploy Regional Multipliers:** Reallocate promotional and advertising budgets away from highly saturated, discount-dependent zones. Redirect those funds toward the geographical areas highlighted on your map that show strong organic demand and a higher concentration of premium, full-price purchases.